

Ravindra Kapse

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EXPERIENCE

Mobius by Gaian Solutions

Hyderabad, Telangana

AI Research Scientist

June 2025 – Present

- Architected an autonomous AI agent for **AI Twinning and Synthetic Data Generation** that builds high-fidelity digital twins through PDE/SPDE numerical simulation, sparse equation discovery (SINDy/WSINDy), and causal DAG learning — generating privacy-guaranteed synthetic data via a ReAct-based agentic loop.
- Designed AI Twinning pipelines combining physics-informed models (FEM, PINNs, neural operators) with data-driven generative models (CTGAN, TimeGAN, diffusion) and causal inference (do-calculus, counterfactual reasoning) for hybrid digital twin construction
- Developed the agent's tool orchestration framework with DAG-scheduled parallel execution, episodic memory for strategy learning, and human-in-the-loop governance gates for compliance-critical Synthetic Data Generation.
- Researched **symbolic AI** systems using causal analysis, counterfactual simulation, and semantic graph rewriting for adaptive agent behavior.
- Investigated **Utility-based Agents** frameworks using game-theoretic reasoning in multi-agent systems(**MAS**).
- Implemented model-agnostic meta-learning (**MAML**) for graph neural networks (**GNN**) to enable rapid adaptation across structured tasks.

Indian Institute of Tropical Meteorology

Pune, Maharashtra

Research Intern

May 2024 – July 2024

- I worked on solving partial differential equations(**PDEs**) using deep learning, specifically with a technique known as Physics-Informed Neural Networks (**PINNs**).
- I gained experience working on High-Performance Computing (**HPC**) systems.

ETH Zurich

Switzerland

Research Project

August 2024

- Utilized Physics-Informed Neural Networks to predict daily terrestrial water storage anomalies by solving water balance equations and **GRACE**-satellite dataset.
- Developed strong research and teamwork abilities.

EDUCATION

Indian Institute of Science, Bangalore, Karnataka

M.Tech, GPA: 8.3 2023 – Jun 2025

University Institute of Technology, Bhopal, M.P.

B.Tech, GPA: 7.36 2017 – Jun 2021

PROJECTS

Building image classification

Developed a machine learning model using YOLOv8n and ResNet50 to detect and classify buildings from Google Street View images, achieving 77.6 % accuracy and securing **2nd place in a Kaggle competition**.

Meesho: Predicting Clothing Attributes

Achieved **8th rank out of 197 teams** in the Meesho Kaggle competition for predicting clothing attributes. Applied advanced techniques including YOLO, ResNet with a multi-head architecture, and XGBoost to develop an accurate and efficient solution.

Chatbot and Feedback Tool

Developed a website-integrated chatbot using Grog API with multi-turn conversation support and Phoenix -based observability. Built a Streamlit feedback tool for annotations and an analytics dashboard for trace export and reporting.

Physics Informed Neural Networks

Worked with PINNs to solve various PDEs, including Burgers', heat, Navier-Stokes, SWE and advection equations. This experience enhanced my computational modeling techniques, allowing for improved accuracy and efficiency in simulating complex physical phenomena.

InstaBot AI

An automated bot that generates prompts and captions using DeepSeek-R1, creates AI images with Flux, and auto-posts them on Instagram, streamlining content creation.

Meta-Learning for Graph Neural Networks

Implemented model-agnostic meta-learning (MAML) using Learn2Learn and PyTorch Geometric to enable fast adaptation on protein graph classification tasks. Trained GNNs to generalize across multiple few-shot learning scenarios using meta-training episodes.

SKILLS & INTERESTS

Programming: Python
Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, PyTorch Geometric, OpenCV, DeepXDE, Learn2Learn, RDFlib, DoWhy, Hugging Face Transformers, vLLM, Ollama, Streamlit
Machine Learning & Deep Learning: Transformers, CNNs (ResNet, VGG), Vision Models (YOLOv8), Graph Neural Networks (GNNs), Meta-Learning (MAML), Physics-Informed Neural Networks (PINNs), Transfer Learning, Model Ensembling, Class Imbalance Handling
Causal Inference & Reasoning: DoWhy, Counterfactual Analysis, Structural Causal Models (SCM), Causal Discovery (basic), Multi-agent Systems, Game Theory
Data Science & Analytics: Data Cleaning, Feature Engineering, EDA, Visualization (Matplotlib, Seaborn), Dashboarding (Streamlit), Statistical Analysis
MLOps & Systems: Git, Linux, HPC/Cluster Computing, Experiment Tracking, Model Deployment (basic APIs), Phoenix Observability
Computer Vision: Image Classification, Object Detection, Image Segmentation, Data Augmentation, Annotation Pipelines
Geospatial & Climate Data: GRACE, ERA5, Satellite Data Processing, Time Series Forecasting, Spatio-temporal Modeling, Hydrological Modeling

AWARDS

- 2024: Winner of the **OLA Thought Paper Contest** for innovative ideas on boosting India’s economic growth.
- 2017: Received the **Mukhyamantri Medhavi Student Award** for excellence in academic performance.

COURSES

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| • Machine Learning for Cyber-Physical Systems | • Mathematics for Engineers |
| • Theory and Applications of Bayesian Learning | • Data Analysis for Earth System Science |
| • Mathematical Methods for Machine Learning | • Scientific Writing and Presentation |